

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Second Semester of M. Tech. Examination (Mechanical, CAD/CAM)

December 2012

ME – 707.01 Computer Aided Manufacturing

Date: 14.12.2012, Friday

Time: 10:00 a.m. to 01:00 p.m.

Maximum Marks: 70

Instructions:

1. Section I and II must be attempted in separate answer sheets.
2. Make suitable assumptions and draw neat figures wherever required.
3. Use of scientific calculator is allowed.

SECTION- I

- Q1 (a) Define Computer Aided Manufacturing & classify its applications into broad categories 3
- (b) Explain the geometrical and motion statements in APT stating an illustration 4
- (c) State and explain the factors which justify the manufacturing of parts on NC/CNC machines. 4
- Q2 (a) State the reasons for using rapid prototyping. What is the basic methodology followed for current Rapid Prototyping techniques? 4
- (b) Differentiate between NC and conventional machine tools. 4
- (c) Explain with sketch 5 axis machining stating illustrations. 4

OR

- Q2 (a) With neat schematic diagram explain selective laser sintering process of rapid prototyping. 4
- (b) Draw a sketch of vertical spindle machining center with a rotary table and specify all the axes on it. State the rules of axes designation followed. 4
- (c) Describe four G (preparatory) functions stating illustrations. 4
- Q3 (a) Write a manual part program for edge milling a low-carbon steel rectangular component of size 120 mm x 80 mm x 10 mm. Select and mention workpiece zero, start point and sketch tool path. 6
- (b) Describe with sketch recirculating ball screw used in CNC machines stating advantages. What is preloading? 3
- (c) What is a canned cycle? Give two examples. 3

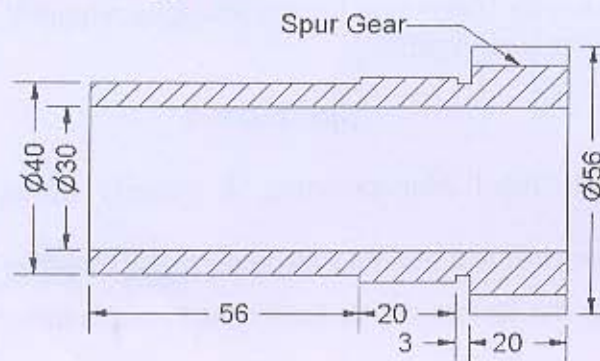
OR

- Q3 (a) Write a manual part programming using canned threading cycle for cutting a metric thread of 0.75 mm pitch on an aluminium rod of $\Phi 15$ mm diameter over length of 25mm. Maximum permissible depth of cut is 0.2mm. 6
- (b) With the help of flowchart, explain the procedure to be followed for generating NC part program using any CAM software. 3
- (c) Explain a drilling canned cycle. Write its statement and show the motions with the help of a sketch. 3

SECTION- II

- Q4 (a)** Discuss the software and hardware elements of CIM. 5
(b) Define Automation? Brief the types of Automation. 3
(c) Differentiate between contact & non contact inspection techniques. 3

- Q5 (a)** Write Opitz code for following product. 3



- (b)** Discuss the Evolution of CIM. 4
(c) What is co-ordinate metrology? Why it is applied? Explain any one inspection instrument using co-ordinate metrology. 5

OR

- Q5 (a)** Differentiate between 4
 (i) AGV and RGV
 (ii) Design Attributes & Manufacturing Attributes
(b) Explain the clustering by Bond Energy algorithm for part family formation with the help of appropriate illustration. 4
(c) With the help of suitable illustrative example, explain three basic structures used in classification and coding schemes for the formation of part families. 4

- Q6 (a)** Write short notes on *any three*: 12
 1. Machine vision system
 2. ASRS
 3. MICLASS
 4. Proximity Sensors

OPITZ Table

Digit 1			Digit 2			Digit 3			Digit 4			Digit 5			
Part class			External shape, external shape elements			Internal shape, internal shape elements			Plane surface machining			Auxiliary holes and gear teeth			
0	Rotational parts	$L/D \leq 0.5$	0	Smooth, no shape elements		0	No hole, no breakthrough		0	No surface machining		0	No auxiliary hole		
1		$0.5 < L/D < 3$	1	Stepped to one end or smooth	No shape elements	1	Smooth or stepped to one end	No shape elements	1	Surface plane and/or curved in one direction, external		1	Axial, not on pitch circle diameter		
2		$L/D \geq 3$	2		Thread	2		External plane surface related by graduation around a circle		2	Axial on pitch circle diameter				
3			3		Functional groove	3		External groove and/or slot		3	Radial, not on pitch circle diameter				
4				4	Stepped to both ends	No shape elements	4	Stepped to both ends	No shape elements	4	External spline (polygon)		4	Axial and/or radial and/or other direction	
5			5	Thread		5	Thread		5	External plane surface and/or slot, external spline		5	Axial and/or radial on pitch circle diameter (p.c.d.) and/or other directions		
6			6	Functional groove		6	Functional groove		6	Internal plane surface and/or slot		6	Spur gear teeth		
7	Nonrotational parts		7	Functional cone		7	Functional cone		7	Internal spline (polygon)		7	Bevel gear teeth		
8			8	Operating thread		8	Operating thread		8	Internal and external polygon, groove and/or slot		8	Other gear teeth		
9			9	All others		9	All others		9	All others		9	All others		